

A Comparative Study of Deep Learning Models and XAI for Breast Ultrasound Image Classification

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Abstract—Breast cancer continues to be a major cause of death among women globally. Early diagnosis greatly enhances the chances of survival, and ultrasound (USG) is commonly utilized for its safety and efficiency, especially in women with dense breast tissue. However, manual interpretation of USG images is time-consuming and subjective. This study presents a deep learning-based approach to automatically classify breast tumors into benign, malignant, or normal categories using USG images. Two public datasets, BUSI and Mendeley, were combined to create a dataset of 1,028 labeled images. Preprocessing steps included checking duplicate, normalization, noise reduction, contrast enhancement, and data augmentation. A total of nine models were evaluated, including six CNN-based architectures (MobileNetV2, MobileNetV3Large, VGG-16, ResNet-50, InceptionV3, EfficientNetB0) and three Transformer-based models (ViT, MobileViT, LeViT). Among all models, MobileViT achieved the highest performance with 93% accuracy, 95% precision, 91% recall, and 92% F1-score. This model outperformed ResNet-50, MobileNetV2, and MobileNetV3Large (each with 86% accuracy), as well as ViT (83%) and LeViT (75%). Grad-CAM visualizations confirmed MobileViT focus on tumor-relevant regions, enhancing interpretability. These results emphasize MobileViT strength in extracting both local and global features from images, making it well-suited for automated breast cancer detection. The integration of Explainable AI also enhances the model's transparency and relevance for clinical use.

Keywords—breast cancer ultrasound, deep learning, transfer learning, transformer, XAI

I. INTRODUCTION

Breast cancer is a highly prevalent and severe cancer, affecting approximately one in eight women worldwide [1]–[2]. The Global Cancer Observatory reported that in 2020, breast cancer accounted for 11.7% of all cancer diagnoses (2.26 million cases) [3], [4], [5]. Its incidence has increased by an average of 0.5% per year over the past four decades [6], with an estimated 297,790 new cases and 43,170 deaths in the United States by 2023 [7]. In China, there were 303,600 new cases and 70,400 deaths in 2015, and these figures are projected to increase by more than 11% by 2030 [8]. In the Republic of Korea, there were 277,523 new cases and 82,688 deaths reported in 2021, an increase of 10.8% from the previous year [9]. The significant number of cases and deaths emphasizes the need for early detection, which may decrease the risk of mortality by up to 4%.[8]–[9].

One of the efficient and affordable early detection methods is ultrasonography (USG), especially for women with dense breast tissue and under the age of 40 years [12], because it has

been proven to significantly reduce the death rate from breast cancer [10]. However, the process of classifying USG images remains complex and requires high expertise. Analysis of complex anatomical structures makes manual classification subjective, because it is highly dependent on the experience and knowledge of individual medical personnel [12]. This manual procedure is also time-consuming because each image must be examined and analyzed thoroughly [13]. The growing adoption of ultrasound (USG) in healthcare settings drives the development of technology to enhance diagnostic speed and accuracy.

With the increasing utilization of ultrasonography (USG) as a safe and effective tool for early detection, the integration of artificial intelligence (AI), particularly deep learning (DL), has gained attention as a promising solution for automated breast tumor classification [12]–[15]. This study applied a deep learning approach to classify breast tumors based on ultrasound images. Compared with other techniques such as mammography which are more expensive and involve radiation, USG offers a safer and more affordable alternative, especially for young women with dense breast tissue [12]. Thus, USG-based classification is considered as a more suitable method to support accurate, non-invasive, and efficient breast cancer diagnosis.

The primary objective of this study is to design an accurate and efficient system for classifying breast tumors, aimed at supporting medical professionals in making well-informed decisions [14]. The system leverages deep learning techniques, specifically employing both Transfer Learning (TL) and Transformer-based models, which are well-known for their strong performance in image classification. In this research, six pre-trained CNN models were implemented using Transfer Learning namely MobileNetV2, MobileNetV3, VGG16, ResNet-50, InceptionV3, and EfficientNetB0 alongside three Transformer-based models: Vision Transformer (ViT), MobileViT, and LeViT. These models were assessed based on their effectiveness in categorizing breast ultrasound images into benign, malignant, and normal classes. A unified dataset of ultrasound images was used to train and evaluate all models, allowing for a thorough comparison between CNN-based transfer learning approaches and Transformer-based architectures. To ensure a balanced and robust evaluation of classification performance, multiple metrics were employed, including accuracy, precision, recall, and F1-score [18]. In addition, visual inspection of the localization capability of each model has

been conducted using Grad-CAM to highlight the regions of interest. The visualized results are then compared across all models to assess their interpretability and effectiveness in localizing tumor areas.

II. RELATED WORK

Breast cancer affects a large number of women and continues to be a major cause of death among women around the world. [19]. Recently, artificial intelligence, particularly deep learning, has gained prominence for its capacity to process vast data, recognize patterns, and aid medical decision-making. A comprehensive review analyzing 200 studies published between 2000 and 2022 highlighted the significant contribution of machine learning techniques in various medical applications, including cancer diagnosis, risk prediction, and treatment outcome estimation demonstrating their potential to improve accuracy and efficiency in digital diagnostic systems [15].

In the field of breast imaging, deep learning particularly Convolutional Neural Networks (CNNs) has been extensively used to detect tumors in both mammography and ultrasound scans [18]–[19]. Due to limited labeled data, Transfer Learning (TL) has become a popular strategy, allowing pre-trained models like InceptionV3, MobileNetV2, and VGG-16 to be adapted for medical tasks [22]. Comparative studies have shown that TL models such as InceptionV3 can significantly outperform custom CNNs and traditional machine learning methods. For instance, InceptionV3 achieved 85.13% accuracy in one study, highlighting its superior performance and efficiency. These results highlight the effectiveness of transfer learning in diagnosing breast cancer from ultrasound images, especially in clinical environments with limited resources. [23].

Transformer-based models like Vision Transformer (ViT), MobileViT, LeViT, and FastViT have recently become popular in medical imaging for their strength in capturing long-range dependencies through self-attention mechanisms [22]–[25]. Although these models have been applied in tasks like skin lesion and retinal disease classification, their use in breast ultrasound remains limited. These models have shown strong potential in multiple medical imaging applications, including the classification of skin lesions and retinal conditions. However, the application of Transformer models in breast ultrasound imaging is still relatively limited. Studies that directly compare the performance of Transformer-based and CNN-based models are also rare, although there are several lightweight Transformer variants designed specifically for medical applications with computational constraints.

Two commonly used datasets for breast ultrasound classification are the BUSI dataset, which includes three classes (benign, malignant, and normal) [28], and the Mendeley dataset, which contains two classes (benign and malignant) [29]. Some studies have combined both datasets but excluded the normal class due to class imbalance, which limits clinical applicability. In real-world screening scenarios, however, the inclusion of the normal class is essential to accurately represent healthy patients. A previous study using VGG16 on the merged dataset reported an accuracy of 91%, sensitivity of 94%, specificity of 88%, and an F1-score of 89% [21]. While these results are encouraging, further improvement is still needed particularly in achieving accuracy above 91% considering that diagnostic precision is critical in

healthcare. Including the normal class is not only necessary for diagnostic completeness but also contributes to building clinically relevant classification models.

Despite significant progress in the development of CNN and Transformer models for medical image classification, there is still a significant research gap regarding the comprehensiveness between these two deep learning paradigms specifically for the task of breast ultrasound image classification. Furthermore, only a few studies have explicitly compared the performance of these two approaches on a combined dataset that includes the clinically important third class (benign, malignant, and normal). Addressing this gap is essential for building diagnostic models that are accurate, dependable, and suitable for clinical use.

In addition to performance concerns, one major drawback of deep learning models is their limited interpretability, commonly known as the “black box” issue. [28]–[29]. This hampers clinical adoption, as medical professionals require understandable and trustworthy predictions. To address this, Grad-CAM is used in this study to visualize attention regions and assess whether models focus on clinically relevant tumor areas.

Thus, this research provides a comparative analysis of Transfer Learning and Transformer-based models for classifying breast ultrasound images as benign, malignant, or normal. Grad-CAM is also applied to examine model interpretability, ensuring strong performance and clinical applicability.

III. METHODOLOGY

Fig. 1 illustrates the flowchart of the proposed method for classifying breast cancer.

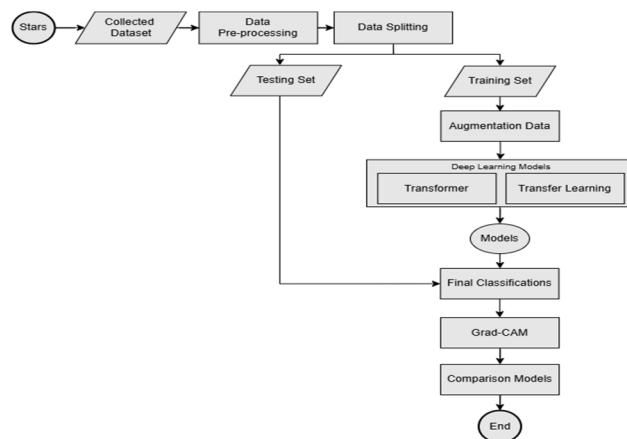


Fig. 1. Methodology for classifying breast ultrasound images using deep learning and Grad-CAM.

A. Collected Dataset

This study employs two publicly accessible breast cancer ultrasound image datasets: the Breast Ultrasound Images (BUSI) dataset [28] and the Mendeley BUS dataset [29]. These datasets are used for two main objectives: first, to expand the size of the training set to reduce the risk of overfitting and bias; and second, to consolidate the classification into three categories: malignant, benign, and normal. The combination of both datasets is intended to evaluate the generalization capability of the proposed classification model. Table I provides the distribution of images from each dataset, while Fig. 2 and Fig. 3 illustrate

sample images from the BUSI and Mendeley datasets, respectively.

TABLE I. DISTRIBUTION OF SAMPLES PER CLASS IN THE BREAST CANCER ULTRASOUND DATASET

Dataset	Benign	Malignant	Normal	Total
Mendeley	100	150	-	250
BUSI	437	210	133	780
Total	563	360	133	1,030

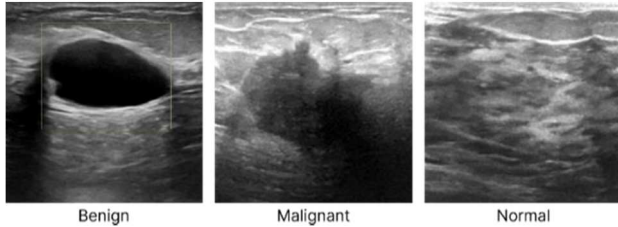


Fig. 2. Sample USG of the BUSI dataset.

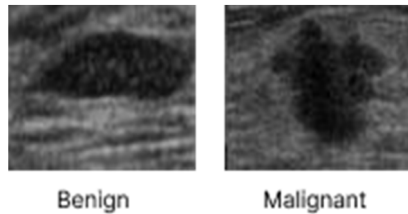


Fig. 3. Sample USG of the Mendeley BUS dataset.

B. Preprocessing

During the preprocessing phase, all images were standardized to JPG format to maintain consistency, and duplicate entries within the BUSI dataset were eliminated to prevent potential bias. The resulting dataset included a total of 1,028 breast ultrasound images, comprising 359 malignant, 536 benign, and 133 normal cases. To reduce speckle noise, a 5×5 median filter was utilized, followed by the application of CLAHE (Contrast Limited Adaptive Histogram Equalization) to boost local contrast and highlight important anatomical details. The images were then resized to $224 \times 224 \times 3$ pixels and normalized to a pixel value range between 0 and 1, ensuring they met the model input criteria and facilitated efficient training convergence.

C. Splitting Data

From the total of 1,028 ultrasound images, the dataset was split into 80% for training and 20% for testing. A total of 104 images were allocated for testing, while the remaining 924 images were used for training the models.

D. Augmentation Data

To overcome the limitations posed by the small dataset size and to mitigate overfitting, data augmentation techniques were employed to introduce variability and enhance the model's ability to generalize. As detailed in Table II, two augmentation strategies were utilized: random horizontal and vertical flipping, as well as random zooming of up to 20%.

TABLE II. DETAIL OF AUGMENTATION TECHNIQUES

Attribute	Value
Randomized Vertical Flip	True

Randomized Horizontal Flip	True
Random Zoom	0.2

E. Deep Learning Models

This research examines ten deep learning models, grouped into two main categories:

- **Transfer Learning Models:** MobileNetV2[32], MobileNetV3[33], VGG16[34], ResNet-50[35], InceptionV3 [36], and EfficientNetB0[37]. These pre-trained CNNs were used to extract deep features and fine-tuned for breast ultrasound classification.
- **Transformer-based Models:** Vision Transformer (ViT)[24], MobileViT[25], and LeViT[26]. These models were chosen for their self-attention capabilities and strong performance in recent medical image analysis.

F. Performance Evaluation Models

To evaluate the performance of both transfer learning and transformer-based models, this study employs metrics derived from the confusion matrix, incorporating values such as True Positive (TP), True Negative (TN), False Positive (FP), and False Negative (FN). These metrics form the basis for calculating key indicators, including Accuracy, Precision, Recall, and F1-Score [38].

$$\text{accuracy} = \frac{TP + TN}{TP + FP + TN + FN} \times 100\% \quad (1)$$

$$\text{precision} = \frac{TP}{TP + FP} \times 100\% \quad (2)$$

$$\text{recall} = \frac{TP}{TP + FN} \times 100\% \quad (3)$$

$$\text{F1-score} = \frac{2 \times \text{precision} \times \text{recall}}{\text{precision} + \text{recall}} \times 100\% \quad (4)$$

IV. EXPERIMENTAL RESULT AND DISCUSSION

A. Environmental Configuration Model Training

All experiments were carried out using the Google Colaboratory platform, which provides GPU acceleration to support deep learning tasks. The training and evaluation of both transfer learning and transformer-based models were conducted in this environment. All breast ultrasound images were resized to 224×224 pixels before being processed by the models. Table III presents the detailed configuration settings used during training and evaluation.

TABLE III. HYPERPARAMETER TUNING IN EXPERIMENT

Model	Image Size	Epoch	Batch Size	Learning Rate	Optimizer	Loss
All Model	224 x 224	30	32	0.0001	Adam	Categorical Cross Entropy

B. Performance Evaluation Result

The training process involved both transfer learning and transformer-based architecture. Table IV presents the performance results on the test dataset, including accuracy (1), precision (2), recall (3), and F1-score (4). Fig. 4 illustrates the confusion matrix, highlighting each model's capability in

distinguishing between benign, malignant, and normal categories.

TABLE IV. RESULT OF QUANTITATIVE PERFORMANCE EVALUATION

Model	Accuracy	Precision	Recall	F1-Score
MobileNetV2	86,54%	85,17%	81,83%	83,12%
MobileNetV3Large	86,54%	88,10%	83,91%	85,72%
InceptionV3	75,96%	84,94%	67,59%	72,16%
ResNet-50	86,54%	86,66%	83,91%	85,09%
VGG-16	72,12%	68,48%	65,43%	66,41%
EfficientNetB0	82,69%	87,96%	77,60%	81,30%
ViT	83,65%	84,89%	78,84%	81,12%
MobileViT	93,27%	95,17%	91,23%	92,99%
LeViT	75,96%	72,65%	79,50%	74,33%

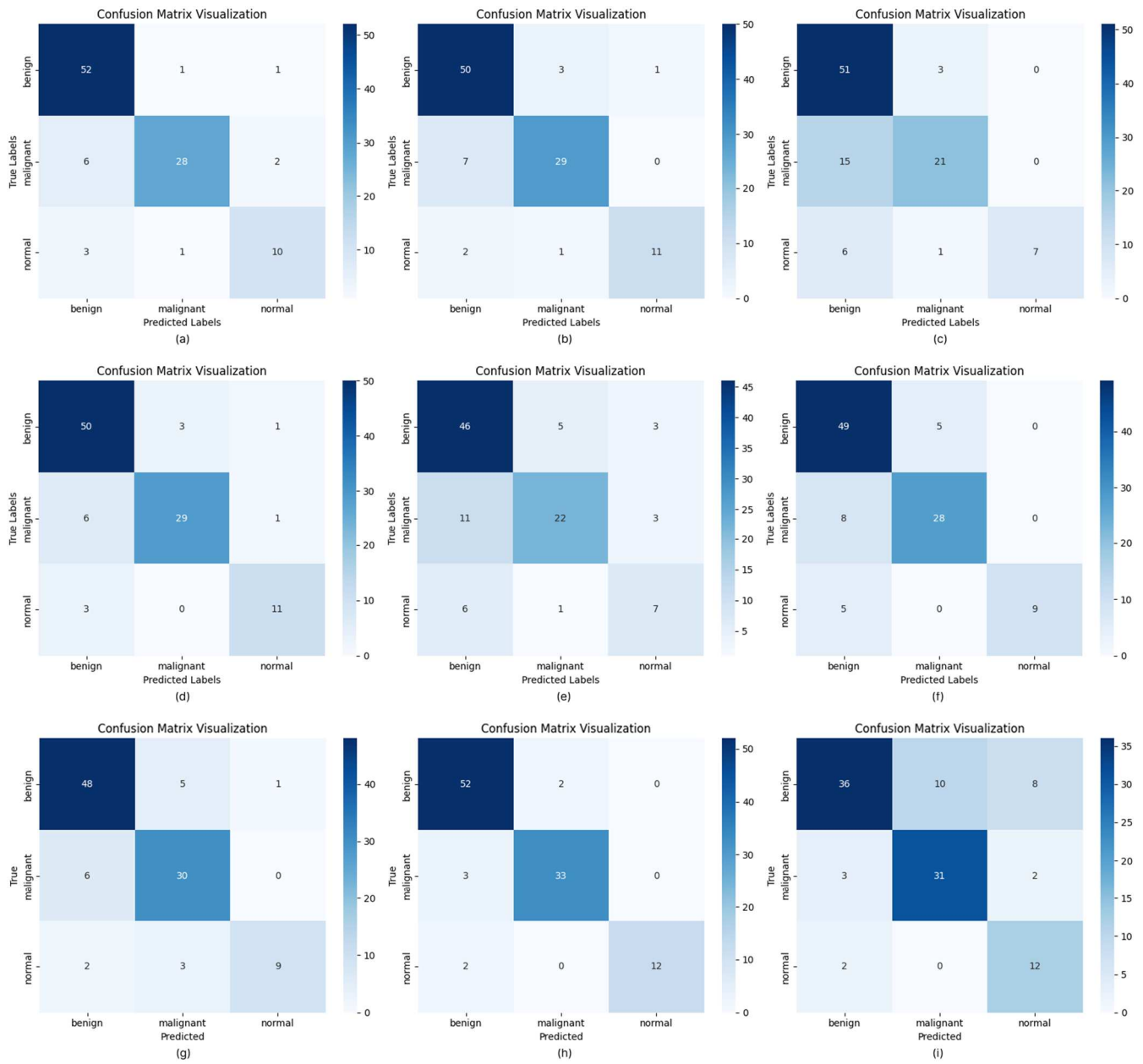


Fig. 4. Confusion matrix for testing result of variois deep learning models a) MobileNetV2. (b) MobileNetV3Large. (c) InceptionV3. (d) ResNet-50. (e) VGG-16. (f) EfficientNetB0. (g) ViT. (h) MobileViT. (i) LeViT.

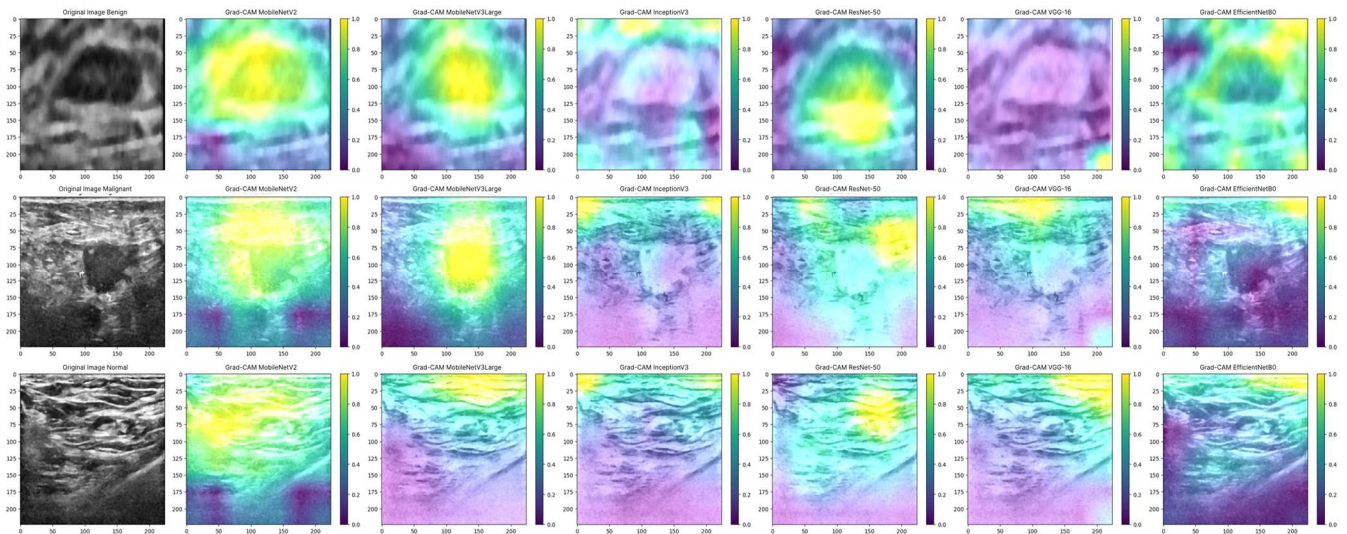


Fig. 5. Grad-CAM visualizations of transfer learning models.

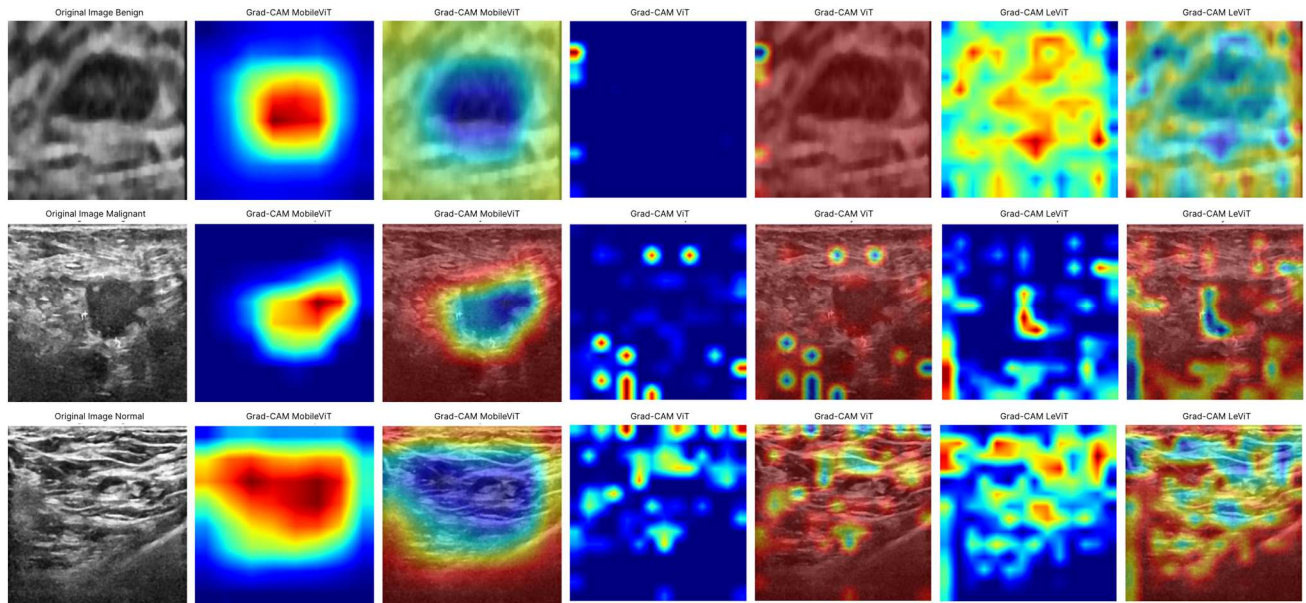


Fig. 6. Grad-CAM visualizations of transformer-based models.

This section outlines the evaluation results of deep learning models used to classify breast ultrasound images. The models include CNN-based transfer learning (MobileNetV2, MobileNetV3Large, ResNet-50, InceptionV3, VGG-16, EfficientNetB0) and transformer-based models (ViT, LeViT, MobileViT), evaluated using four metrics: accuracy, precision, recall, and F1-score. As presented in Table IV, MobileViT recorded the top performance with 93% accuracy, 95% precision, 91% recall, and 92% F1-score. These results demonstrate the model's strong classification performance, including high sensitivity and specificity an essential factor in clinical diagnostics where minimizing false positives and false negatives is crucial. MobileViT's superior results are attributed to its hybrid architecture, which combines the strengths of convolutional layers in capturing local features with transformers' ability to model global context, effectively addressing the complex nature of ultrasound images.

Other lightweight transfer learning CNNs such as MobileNetV2, MobileNetV3Large, and ResNet-50 performed competitively, each achieving 86% accuracy with balanced precision and recall between 81% and 88%. These models demonstrate reliable performance but with slightly lower sensitivity than MobileViT, which is a concern in the medical domain due to the high cost of false negatives. EfficientNetB0 followed closely at 82% accuracy, balancing computational efficiency and classification capability. In contrast, InceptionV3 and VGG-16 showed lower accuracy (75% and 72%) and notably poor recall (67% and 65%, respectively), reflecting limitations in generalization and feature extraction on this dataset.

Among transformer-based models, ViT achieved 83% accuracy with high precision (84%) but lower recall (78%), suggesting a more conservative prediction strategy that risks overlooking malignant cases. LeViT, despite its efficient architecture, delivered suboptimal results (75% accuracy,

74% F1-score) and showed instability in class predictions. These results align with prior research suggesting that transformer-based models often face challenges when applied to limited or specialized datasets, particularly in the absence of substantial pretraining or comprehensive data augmentation.

The confusion matrix in Fig. 4 further confirm these observations. MobileViT's matrix (Fig. 4h) reveals strong performance in detecting malignant cases (33 correctly classified, only 3 misclassified), supporting its high recall. In contrast, VGG-16 and LeViT (Fig. 4e and 4i) show significant misclassification between benign and malignant categories, highlighting their diagnostic limitations. ViT's confusion matrix (Fig. 4g) is more balanced but still less robust than MobileViT in the malignant class.

To complement these quantitative results, Grad-CAM visualizations were used to assess interpretability. As shown in Fig. 5 and Fig. 6, MobileViT produced the most focused and clinically relevant attention maps, with heatmaps accurately localizing tumor regions and avoiding background noise. This visual interpretability supports the model's strong performance and demonstrates its capability to focus on clinically relevant regions. In comparison, ViT produced broader and less concentrated activations, and LeViT often focused on irrelevant regions, which may explain their lower classification reliability.

Among the transfer learning CNNs, ResNet-50 and EfficientNetB0 provided relatively accurate visual attention, though occasionally highlighting surrounding tissues. MobileNetV2 and MobileNetV3Large showed moderate attention consistency, while VGG-16 and InceptionV3 produced scattered, ambiguous heatmaps, reflecting their weaker classification outcomes. These qualitative differences highlight the importance of not only high accuracy but also model transparency, particularly in medical applications where clinicians must understand and trust AI decisions.

V. CONCLUSION

This research conducts a comparative evaluation of deep learning models and explainable AI (XAI) techniques for classifying breast ultrasound images into three categories: benign, malignant, and normal. The analysis covers both CNN-based transfer learning models and transformer-based architectures, focusing on both classification accuracy and interpretability using Grad-CAM visual explanations.

Among all models, MobileViT achieved the highest overall performance, with 93% accuracy, 95% precision, 91% recall, and 92% F1-score. Its Grad-CAM heatmaps effectively highlighted tumor regions with high clinical relevance, demonstrating the value of combining predictive strength with visual explanation. CNN models such as MobileNetV2, MobileNetV3Large, and ResNet-50 also performed well (each with 86% accuracy), while InceptionV3 (75%) and VGG16 (72%) showed lower performance and recall, increasing the risk of false negatives. Transformer models like ViT (83%) and LeViT (75%) produced less stable results and less focused Grad-CAM outputs, indicating weaker interpretability.

The use of Grad-CAM offers meaningful understanding of how each model makes its predictions, highlighting the critical role of integrating both accuracy and interpretability in clinical AI applications. Future work will focus on developing

hybrid architectures that integrate the strengths of CNNs and transformers building on the success of MobileViT to improve diagnostic performance. Additionally, advanced XAI techniques such as SHAP and LIME will be explored to offer deeper and more actionable insights for real-world clinical use.

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